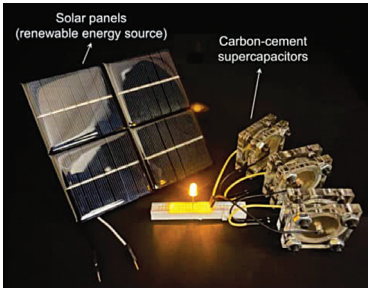




45gm, is flexible and soft. Made from affordable and accessible materials, it has the potential for large-scale production. In preliminary testing, the robot navigated horizontally at 10mm/s and vertically at 0.2mm/s.

Supercapacitors developed from ancient materials

MIT researchers have developed a supercapacitor that has the potential to be integrated into a home's concrete foundation, where it could store an entire day's energy without significantly increasing its cost while maintaining its structural integrity. The team created supercapacitors from cement mixed with carbon black, forming conductive structures with a high internal surface area. The carbon structures accumulate



Supercapacitors developed from ancient materials (Courtesy of the researchers)

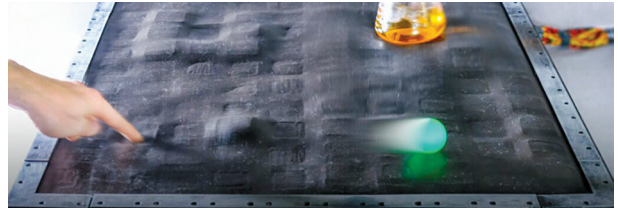
charges, creating a potent supercapacitor. A 45 cubic metre block of this material can store around 10 kilowatt-hours, suitable for daily home energy use. These supercapacitors charge and discharge faster than batteries and could store energy from renewable sources. After crafting small 1-volt supercapacitors, the team aims to create larger versions for household power storage, having found a balance between the material's storage capacity and structural strength.

Metasurface breakthrough boosts advanced telecommunications

Researchers at Johns Hopkins University have developed an innovative compact and lightweight reflective surface that has the potential to transform communications in densely populated areas, offering unparalleled management of electromagnetic signals, including radio waves. The team analysed a two-layer metasurface, incorporating various elements to enhance control. This design created a versatile cascaded metasurface that manages magnitude and phase, fitted on an economical printed circuit board. The material corrects inconsistent signal loss using two resonant materials, resulting in a compact, lightweight metasurface for advanced electromagnetic control, potentially boosting cell or Wi-Fi signals around a city. The team's dynamic cascaded metasurface bounces signals off patterned surfaces to enhance them. The researchers believe this technology could be powered by something as simple as a nine-volt battery.

The future of 3D digital displays

Engineers from the University of Colorado Boulder have developed a unique shape-shifting display. The team has introduced hydraulically amplified self-healing electrostatic (HASEL) actuators in the development of soft robotics.



A new, shape-shifting display can sense and respond to human touch (Credit: Brian Johnson)

Though not market-ready, these actuators have potential applications in sensory gloves for immersive gaming or conveyor belts that can sort items like apples from bananas. The design, similar to a Scrabble board, includes a 100-square grid of HASEL actuators. These miniature accordion-shaped pouches, filled with fluid, bulge and rise when electrified. They have soft, magnetic sensors responsive to touch, allowing sketching on the surface with a magnetic wand. The team is working to miniaturise the actuators to increase the display's resolution, with potential applications like transforming articles into Braille on a phone screen.

RoboMapper: The future of efficient semiconductor research

Researchers at North Carolina State University have created a robot called RoboMapper, designed to conduct experiments more efficiently and environmentally-friendly, aiding in the discovery of new semiconductor materials with desired properties. The researchers noted that RoboMapper, by compactly placing materials on one chip, reduced greenhouse gas emissions tenfold. Defined by their crystal structure, perovskites surpass silicon in light absorption, making perovskite solar cells more compact and lightweight without compromising conversion capacity. The team used RoboMapper to create and analyse 150 alloys for tandem solar cells, assessing their crystalline structure, optical properties, and light stability. A computational model identified an optimal alloy composition, which was then created through RoboMapper and traditional lab methods for comparison.

Battery design exceeds fast charge targets for lithium cycling

Researchers at Maryland Energy Innovation Institute (MEI2) and University of Maryland (UMD) have developed a mixed